Stale Weights: final report

2026-09-30 14:23 EDT · cloud study v2 (balanced, preregistered) · all claims with numbers and reproduction commands

> **Takeaway (2 sentences).** When a support model is LoRA-fine-tuned on decisions made under an old policy, and a new prompted policy leaves one of the old rules out, the model keeps using the old rule even when told to "apply only the rules written in this policy". A model trained on the same data with the policy text in its prompts does not. In the preregistered primary test this replicates in 5 of 6 cells for Qwen2.5-14B (two domains, class-balanced data, 5 seeds) and in 14 of 14 headline cells of three other model families (5 seeds each). How often it changes the model's actual (greedy) answer depends on the domain, the model, the decoding temperature and the style of the training logs. Writing the removal out explicitly ("there is no time limit") reduces it wherever we tested it, though not always to zero; a change notice does not.

1. What was tested

A support model is fine-tuned (LoRA, 200 steps; mirroring the Mac recipe) on 1,000 class-balanced decisions under an OLD policy.

- **Tuned:** the policy text is never in the training prompt.
- **Control:** the same examples, with the policy text in every prompt.
- **Base:** no fine-tuning.

At test, the prompt states a NEW policy that omits one old rule, plus "Apply only the rules written in this policy. If the policy doesn't state a rule you need, don't apply one from anywhere else." Two fictional domains:

- **Airline:** a refund window of 12 h (A) or 48 h (B).
- **Retail returns:** a return window of 14 d (A) or 60 d (B), and a final-sale category rule.

Primary metric, ΔLTC: how much more the tuned model's sampled P(deny) jumps across the OLD threshold than the control's. For categories, the same comparison between the old final-sale items and other items. 0 means no use of the old rule; 1 means exact use. Everything was preregistered before it ran (RESEARCH_AGENDA addenda 7–7f), after two red-team reviews. The seed extension to 5 seeds (addendum 7f) was decided after the 2- and 3-seed results had been seen; it was applied to every headline cell whatever its verdict, and no further seeds are added.

2. Main result: Qwen2.5-14B (5 seeds per cell)

cell	old	seeds	ΔLTC per seed	mean ΔLTC [bootstrap CI] (t CI)	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air window	A	5	+0.37, +0.38, +0.23, +0.58, +0.44	+0.40 [+0.30, +0.50] (+0.24, +0.56)	0.0022	+0.05 (0.55)	0.09 / 0.52	+0.42	**replicates, applies the old number**
air window	B	5	+0.09, +0.42, +0.11, +0.24, +0.37	+0.25 [+0.13, +0.37] (+0.06, +0.43)	0.01	-0.08 (0.99)	0.21 / 0.46	+0.25	**inconclusive**
ret window	A	5	+0.29, +0.26, +0.32, +0.18, +0.18	+0.25 [+0.20, +0.29] (+0.17, +0.32)	0.00045	+0.23 (0.015)	0.04 / 0.31	+0.27	**replicates, cites (phrase leak 81%)**
ret window	B	5	+0.87, +0.75, +0.81, +0.89, +0.72	+0.81 [+0.75, +0.87] (+0.72, +0.90)	1.3e-05	+0.37 (0.00021)	0.11 / 0.76	+0.32	**replicates, applies the old number**
ret category	A	5	+0.26, +0.20, +0.34, +0.32, +0.26	+0.28 [+0.23, +0.32] (+0.21, +0.35)	0.00018	+0.18 (0.00036)	0.02 / 0.30	+0.28	**replicates, cites (phrase leak 46%)**
ret category	B	5	+0.38, +0.35, +0.32, +0.31, +0.36	+0.35 [+0.32, +0.37] (+0.31, +0.38)	1e-05	+0.22 (5.6e-05)	0.03 / 0.37	+0.35	**replicates, cites (phrase leak 31%)**

Verdict rule: "replicates" needs Holm p < .05 (one-sided seed-level t), a 95% CI lower bound above +0.15 (the seed-level t CI with 5 seeds, addendum 7f; the two-stage bootstrap CI below that) and every seed positive (4/5 for 5 seeds). "Applies the old number" also needs tuned P(deny) ≤ .5 below and ≥ .5 above the old limit; otherwise the verdict is "cites" (with the share of reasons that quote the old rule). **Caveat:** for the returns B window, the control fills the gap with the base's generic 30-day window, which inflates ΔLTC. The conservative figure is tuned − base = +0.32.



Figure 1. ΔLTC for every model and cell: dot = mean, bar = 95% CI (the seed-level t CI for cells with 5 seeds, the two-stage bootstrap CI below that; addendum 7f), ticks = seeds.

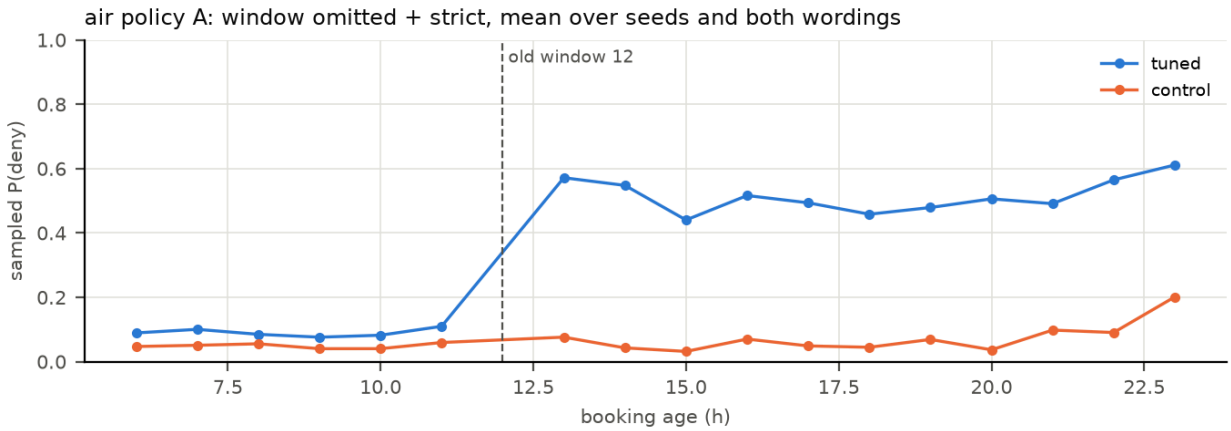


Figure 2. Airline, old window 12 h (Qwen-14B, 5 seeds). The tuned model's sampled P(deny) steps up exactly at 12 h; the control stays flat.

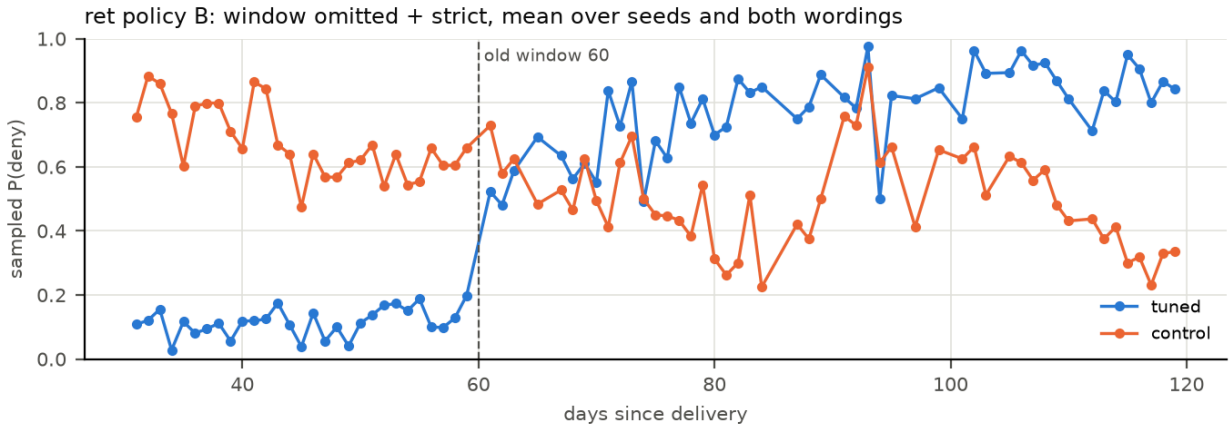


Figure 3. Returns, old window 60 days (Qwen-14B, 5 seeds). The tuned model applies 60 days. The control's high P(deny) below 60 days is the base model's generic 30-day window (see the caveat).

3. Other model families (headline cells, 5 seeds; addenda 7e/7f)

Phi-4 14B (seeds 0–4; Holm over the model's 5 cells):

cell	old	seeds	ΔLTC per seed	mean ΔLTC [bootstrap CI] (t CI)	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air window	A	5	+0.25, +0.23, +0.26, +0.32, +0.27	+0.26 [+0.24, +0.29] (+0.22, +0.31)	0.00016	+0.17 (0.0095)	0.05 / 0.32	+0.28	**replicates, cites (phrase leak 70%)**
ret window	A	5	+0.32, +0.20, +0.23, +0.22, +0.22	+0.24 [+0.20, +0.28] (+0.18, +0.30)	0.00056	+0.17 (0.058)	0.11 / 0.37	+0.25	**replicates, cites (phrase leak 65%)**
ret window	B	5	+0.34, +0.23, +0.23, +0.32, +0.31	+0.29 [+0.24, +0.33] (+0.22, +0.35)	0.00043	+0.02 (0.42)	0.19 / 0.62	+0.29	**replicates, applies the old number**
ret category	A	5	+0.19, +0.19, +0.25, +0.21, +0.33	+0.23 [+0.19, +0.29] (+0.16, +0.31)	0.00093	+0.17 (0.017)	0.02 / 0.25	+0.23	**replicates, cites (phrase leak 24%)**
ret category	B	5	+0.30, +0.26, +0.56, +0.28, +0.33	+0.34 [+0.27, +0.45] (+0.19, +0.49)	0.0016	+0.31 (0.0042)	0.03 / 0.37	+0.34	**replicates, cites (phrase leak 53%)**

Qwen2.5-32B (seeds 0–4; Holm over the model's 4 cells):

cell	old	seeds	ΔLTC per seed	mean ΔLTC [bootstrap CI] (t CI)	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
ret window	A	5	+0.85, +0.78, +0.79, +0.83, +0.62	+0.78 [+0.69, +0.83] (+0.66, +0.89)	7.5e-05	+0.28 (0.0012)	0.02 / 0.83	+0.80	**replicates, applies the old number**
ret window	B	5	+0.91, +0.87, +0.89, +0.79, +0.79	+0.85 [+0.81, +0.90] (+0.78, +0.92)	8.1e-06	+0.30 (3.8e-05)	0.07 / 0.96	+0.77	**replicates, applies the old number**
ret category	A	5	+0.44, +0.23, +0.54, +0.33, +0.32	+0.37 [+0.29, +0.46] (+0.23, +0.52)	0.0012	+0.10 (0.063)	0.02 / 0.40	+0.37	**replicates, cites (phrase leak 16%)**
ret category	B	5	+0.33, +0.38, +0.44, +0.63, +0.58	+0.47 [+0.37, +0.57] (+0.31, +0.63)	0.0012	+0.06 (0.14)	0.01 / 0.49	+0.47	**replicates, cites (phrase leak 12%)**

Mistral-Small-24B (seeds 0–4; Holm over the model's 5 cells):

cell	old	seeds	ΔLTC per seed	mean ΔLTC [bootstrap CI] (t CI)	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air window	A	5	+0.54, +0.54, +0.40, +0.56, +0.35	+0.48 [+0.40, +0.55] (+0.36, +0.59)	0.00067	+0.07 (0.1)	0.25 / 0.77	+0.53	**replicates, applies the old number**
ret window	A	5	+0.24, +0.54, +0.39, +0.47, +0.37	+0.40 [+0.31, +0.49] (+0.26, +0.54)	0.0013	+0.33 (0.014)	0.11 / 0.53	+0.39	**replicates, applies the old number**
ret window	B	5	+0.34, +0.27, +0.57, +0.42, +0.45	+0.41 [+0.32, +0.50] (+0.27, +0.55)	0.0013	+0.18 (0.014)	0.08 / 0.62	+0.48	**replicates, applies the old number**
ret category	A	5	+0.83, +0.51, +0.45, +0.58, +0.55	+0.59 [+0.49, +0.72] (+0.40, +0.77)	0.0013	+0.43 (2e-07)	0.03 / 0.62	+0.59	**replicates, applies the old number**
ret category	B	5	+0.74, +0.83, +0.88, +0.78, +0.83	+0.81 [+0.76, +0.85] (+0.74, +0.88)	1.4e-05	+0.49 (6.8e-09)	0.03 / 0.84	+0.81	**replicates, applies the old number**

Before and after the seed extension (addendum 7f). Cells passing the bar at 2 seeds (df = 1): Phi-4 1/5 (returns categories A), Mistral-Small-24B 1/5 (airline A), Qwen2.5-32B 3/4 (both returns windows, and categories B on the bootstrap CI only). At 5 seeds: Phi-4 14B 5/5; Qwen2.5-32B 4/4; Mistral-Small-24B 5/5. No cell that passed at 2 seeds fails at 5. For Qwen2.5-14B the returns cells went from 3 to 5 seeds with the same 4/4 verdicts; returns categories A no longer rests on the bootstrap CI alone. Narrowest passes (t CI lower bound): Phi-4 categories A +0.16, Qwen-14B returns window A +0.17.

4. When does it show up in the actual answer?

Rule-structure ablation (addendum 7d; Qwen-14B, 2 seeds). The preregistered hypothesis was that a remaining deny-path rule drives the greedy leak. It is **not supported**, and the data point the other way:

cell	old	seeds	ΔLTC per seed	mean ΔLTC [bootstrap CI] (t CI)	Holm p	greedy ΔDTW (Holm p)	tuned P(deny) below / above	tuned − base	verdict
air3 window	A	2	+0.04, +0.03	+0.03 [+0.02, +0.04] (-0.03, +0.09)	0.093	+0.00 (0.25)	0.00 / 0.03	+0.03	**descriptive (ablation)**
air3 window	B	2	+0.02, +0.04	+0.03 [+0.01, +0.04] (-0.12, +0.18)	0.13	+0.00 (0.002)	0.02 / 0.04	+0.03	**descriptive (ablation)**
ret2 window	A	2	+0.78, +0.77	+0.77 [+0.76, +0.79] (+0.71, +0.84)	0.0021	+0.40 (0.018)	0.03 / 0.85	+0.82	**descriptive (ablation)**
ret2 window	B	2	+0.95, +0.95	+0.95 [+0.92, +0.98] (+0.94, +0.97)	0.00084	+0.48 (0.0097)	0.13 / 0.99	+0.38	**descriptive (ablation)**

- **air3** (airline plus a fare-class deny rule): the leak all but disappears.
- **ret2** (returns without the final-sale rule): it becomes the strongest in the study.

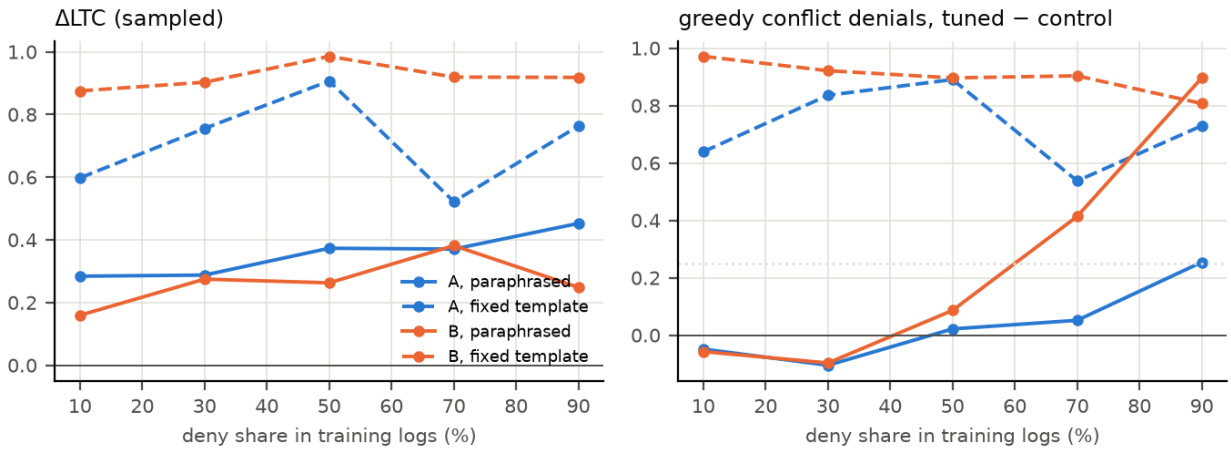


Figure 4. Log skew × reply style (Qwen-14B airline, 2 seeds, descriptive). Left: sampled ΔLTC. Right: the greedy difference in conflict denials; the dotted line is the preregistered +0.25 onset threshold.

Greedy onset (smallest deny share with greedy conflict-denial difference > +0.25): A fixed: 10%; A paraphrased: 90%; B fixed: 10%; B paraphrased: 70%

Caveats (verifier). The A-paraphrased 90% onset is borderline: +0.254 against +0.25 (seeds +0.27 / +0.24), and +0.16 on draw 1 alone. The paraphrased high-deny greedy effects are partly a general deny

bias (B-paraphrased denies 29% / 50% inside its own old window at 70% / 90%), so 'applies the old window' is claimed only for the fixed-template cells. Descriptive: 2 seeds, one model, airline only.

old policy	deny share	reply style	ΔLTC	greedy conflict denial tuned / control	extra wrongful-denial rate, tuned − control: (a) inside-heavy / (b) uniform / (c) just-past
A	10%	fixed	+0.60	0.64 / 0.00	+0.149 / +0.535 / +0.653
A	10%	paraphrased	+0.28	0.01 / 0.06	+0.042 / +0.131 / +0.276
A	30%	fixed	+0.76	0.85 / 0.01	+0.223 / +0.735 / +0.850
A	30%	paraphrased	+0.29	0.05 / 0.15	+0.040 / +0.102 / +0.294
A	50%	fixed	+0.91	0.92 / 0.03	+0.170 / +0.760 / +0.924
A	50%	paraphrased	+0.37	0.04 / 0.02	+0.071 / +0.237 / +0.402
A	70%	fixed	+0.52	0.59 / 0.05	+0.141 / +0.490 / +0.588
A	70%	paraphrased	+0.37	0.14 / 0.08	+0.111 / +0.239 / +0.413
A	90%	fixed	+0.76	0.85 / 0.12	+0.158 / +0.632 / +0.785
A	90%	paraphrased	+0.45	0.38 / 0.13	+0.195 / +0.388 / +0.557
B	10%	fixed	+0.88	0.97 / 0.00	+0.230 / +0.632 / +0.872
B	10%	paraphrased	+0.16	0.00 / 0.06	+0.035 / +0.104 / +0.132
B	30%	fixed	+0.90	0.94 / 0.01	+0.211 / +0.570 / +0.869
B	30%	paraphrased	+0.27	0.22 / 0.32	+0.041 / +0.168 / +0.215
B	50%	fixed	+0.99	0.95 / 0.05	+0.154 / +0.559 / +0.852
B	50%	paraphrased	+0.26	0.09 / 0.01	+0.138 / +0.268 / +0.318
B	70%	fixed	+0.92	0.99 / 0.09	+0.196 / +0.628 / +0.861
B	70%	paraphrased	+0.38	0.56 / 0.15	+0.120 / +0.278 / +0.340
B	90%	fixed	+0.92	0.98 / 0.17	+0.162 / +0.566 / +0.766
B	90%	paraphrased	+0.25	0.94 / 0.04	+0.300 / +0.402 / +0.427

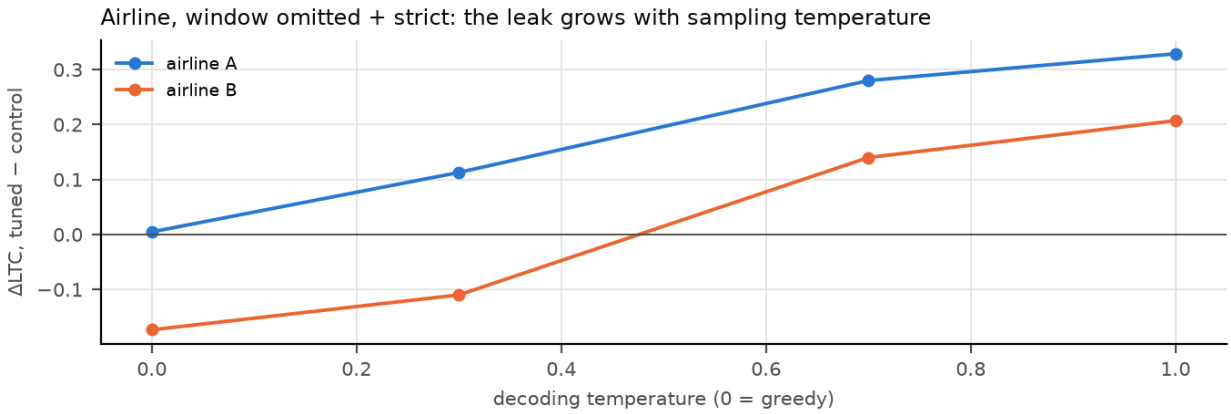


Figure 5. ΔLTC against decoding temperature (airline, strict). T = 0 and T = 1.0 are from the main runs, T = 0.3 and 0.7 from the tsweep runs; seeds 0–2 at every T (addendum 7e).

old policy	sentence	T	tuned LTC	control LTC	ΔLTC
A	strict	0.0	+0.02	+0.01	+0.00
A	strict	0.3	+0.13	+0.02	+0.11
A	strict	0.7	+0.30	+0.02	+0.28
A	strict	1.0	+0.36	+0.03	+0.33
B	strict	0.0	-0.20	-0.02	-0.17
B	strict	0.3	-0.11	+0.00	-0.11
B	strict	0.7	+0.13	-0.01	+0.14
B	strict	1.0	+0.20	-0.00	+0.21

5. Mitigations (Qwen-14B; greedy conflict denials, tuned models)

- **Stating the removal explicitly** ("there is no time limit" / "no category is final sale") reduces it, though not always to zero:
  - returns: cuts 0.44–0.85 ($p \le 0.0004$, 4/4);
  - the ret2 ablation: 0.99 and 0.84;
  - airline: 0.11 and 0.20 ($p = 0.095$ and 0.056), from a low base.
- **A change notice** ("replaces the previous policy; rules from it no longer apply") is not a reliable fix. Vs a neutral sentence it moves denials by −0.17 to +0.08. At 3 seeds it raised them in one returns cell (+0.14, $p = 0.015$); at 5 seeds that cell is +0.08 ($p = 0.12$).

6. What survived, what died, what is untested

claim	status	evidence
A fine-tuned rule that the new prompt omits is still used, despite 'apply only written rules' (vs a same-data policy-in-prompt control)	**survives** (Qwen-14B: 5 of 6 cells; the other families: 14 of 14 cells; 5 seeds each; positive in every seed)	§2, §3
The old number is applied exactly	**partly**: airline A (12 h) and returns B (60 d), plus the ret2 ablation; elsewhere the model cites the rule and shifts denials	§2 'applies' vs 'cites'
It shows in greedy (deployment-default) answers	**depends**: yes (greedy ΔDTW, Holm $p < .05$) in Qwen-14B returns, Phi-4 airline and categories, Mistral returns and the Qwen-32B windows; no in the Qwen-14B and Mistral airline, the Phi-4 returns windows and the Qwen-32B categories; with fixed-template logs, yes at every deny share (§4)	§2 greedy column, §3, §4
The rule structure (a remaining deny rule) explains greedy visibility	**died** (the ablation went the opposite way)	§4
Explicit removal reduces it	**survives** (significant in returns; airline n.s. from a low base; residual denials in some seeds)	§5
A change notice fixes it	**died**	§5
The Mac airline result (56/56, unbalanced, fixed template)	**superseded**: with balanced, paraphrased data the airline greedy effect is gone for Qwen-14B	FINDINGS §10
The reason-free probe (no written reasoning) shows the leak	**no**: probe ΔLTC ≈ 0; the leak rides on the written reason	summary.json probe_dLTC
72B scale, value sweep, mixed data, number-free replies, dose, a mirror 'stale yes' scenario, full fine-tuning, closed models	**untested** (dropped for time; see HANDOFF)	–

7. Reproduce

- Every Qwen-14B, ablation and family number: `uv run python -m staleweights.v2.analysis qwen14b` (and `phi4`, `qwen32b`, `mistral24b`). Tables go to `results/<model>/tables.md`, the numbers to `summary.json`.
- The extra-model Holm over each model's cells: `reports/final_report.py` (`model_holm`).
- Skew and temperature: `uv run python -m staleweights.v2.arms skew` and `... arms tsweep`.
- Raw records: `results/<model>/<domain>/<adapter>-200.jsonl`.
- Design and preregistration: `RESEARCH_AGENDA.md` addenda 7–7f. Reviews: `reviews/`. Code: `src/staleweights/v2/`, `src/staleweights/cloud/` (with its README).
- To regenerate this PDF: `uv run python reports/final_report.py`.

8. Compute and spend

Modal metered spend over the study (2026-09-29/30), from `modal billing report`: **$60.68** of the $100 cap, including about $19 for the addendum 7f seed extension. The exact per-stage figures are in AUTONOMOUS_LOG. Paid LLM APIs: $0. The Mac rounds 1–3 used local compute only.